

```
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import SimpleRNN, Dense
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt
```

```
# Load Dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>
11490434/11490434 ————— **0s** 0us/step

```
# Normalize Data
X_train = X_train / 255.0
X_test = X_test / 255.0
```

```
# Build RNN Model
model = Sequential()
```

```
# RNN Layer
model.add(SimpleRNN(128, activation='relu', input_shape=(28, 28)))
```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not pass an `input`
 super().__init__(**kwargs)

```
# Output Layer
model.add(Dense(10, activation='softmax'))
```

```
# Compile Model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

```
# Train Model
history = model.fit(
    X_train,
    y_train,
    epochs=5,
    validation_data=(X_test, y_test)
)
```

```
Epoch 1/5
1875/1875 ————— 20s 10ms/step - accuracy: 0.8491 - loss: 0.4685 - val_accuracy: 0.9367 - val_
Epoch 2/5
1875/1875 ————— 20s 9ms/step - accuracy: 0.9416 - loss: 0.2034 - val_accuracy: 0.9494 - val_
Epoch 3/5
1875/1875 ————— 19s 10ms/step - accuracy: 0.9538 - loss: 0.1631 - val_accuracy: 0.9650 - val_
Epoch 4/5
1875/1875 ————— 17s 9ms/step - accuracy: 0.9587 - loss: 0.1464 - val_accuracy: 0.9587 - val_
Epoch 5/5
1875/1875 ————— 18s 9ms/step - accuracy: 0.9632 - loss: 0.1334 - val_accuracy: 0.9545 - val_
```

```
# Evaluate Model
test_loss, test_acc = model.evaluate(X_test, y_test)
```

313/313 ————— **1s** 4ms/step - accuracy: 0.9545 - loss: 0.1557

```
print("Test Accuracy:", test_acc)
```

Test Accuracy: 0.9545000195503235

```
# Plot Accuracy
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.show()
```

